

The Comprehensive Finance Reference Guide

A Complete Handbook for Financial Concepts, Markets, and Instruments

Document Purpose: This reference guide covers foundational to advanced financial concepts including capital markets, corporate finance, risk management, derivatives, portfolio theory, and macroeconomic indicators. Designed for practitioners, analysts, and learners seeking a structured deep-dive into modern finance.

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1. Financial Markets Overview

1.1 What is a Financial Market?

A **financial market** is a mechanism that allows buyers and sellers to trade financial instruments such as stocks, bonds, currencies, and derivatives. Financial markets serve several critical economic functions:

- **Price Discovery:** Markets aggregate information from millions of participants to establish fair prices for assets.
- **Liquidity Provision:** Investors can convert assets to cash quickly and at predictable prices.
- **Capital Allocation:** Markets channel savings from households and institutions to productive investments.
- **Risk Transfer:** Through derivatives and insurance products, financial markets allow risk to be redistributed to those best able to bear it.
- **Information Aggregation:** Prices reflect the collective beliefs of all market participants about future cash flows.

1.2 Types of Financial Markets

Primary vs. Secondary Markets

Feature	Primary Market	Secondary Market
Definition	Where new securities are first issued	Where previously issued securities are traded
Participants	Issuers (companies/governments) + investors	Buyers and sellers (investors)
Proceeds go to	The issuing company/entity	The selling investor
Examples	IPOs, Bond Issuances, Rights Issues	NYSE, NASDAQ, BSE, NSE
Price Setting	Fixed or book-built	Determined by supply and demand

Capital Markets vs. Money Markets

Capital Markets deal in long-term instruments (maturity > 1 year):

- Equity (stocks)
- Long-term bonds
- Mortgage-backed securities

Money Markets deal in short-term instruments (maturity $\leq$ 1 year):

- Treasury Bills (T-Bills)
- Commercial Paper (CP)
- Certificates of Deposit (CDs)
- Repurchase Agreements (Repos)
- Federal Funds

1.3 Market Participants

Buy-Side Institutions (those who buy securities):

- Mutual Funds
- Pension Funds
- Hedge Funds
- Insurance Companies
- Sovereign Wealth Funds
- Retail Investors

Sell-Side Institutions (those who facilitate transactions):

- Investment Banks (Goldman Sachs, Morgan Stanley, JPMorgan)
- Broker-Dealers
- Market Makers
- Commercial Banks

Regulatory Bodies:

- **USA:** SEC (Securities and Exchange Commission), CFTC, FINRA, Federal Reserve
- **India:** SEBI (Securities and Exchange Board of India), RBI, IRDAI
- **UK:** FCA (Financial Conduct Authority), Bank of England
- **EU:** ESMA (European Securities and Markets Authority), ECB

1.4 Market Efficiency — The Efficient Market Hypothesis (EMH)

Developed by Eugene Fama in 1970, the **Efficient Market Hypothesis** states that asset prices fully reflect all available information at any given time. EMH has three forms:

Form	Information Reflected	Implication
Weak Form	All past trading prices and volumes	Technical analysis cannot produce alpha
Semi-Strong Form	All publicly available information	Fundamental analysis cannot produce sustained alpha
Strong Form	All information (public + private)	Even insider information is priced in

Challenges to EMH:

- Anomalies like the January Effect, momentum, and value premiums
- Bubbles and crashes (Dot-com 2000, Housing 2008)
- Behavioral finance evidence

2. Time Value of Money

2.1 The Core Principle

The **Time Value of Money (TVM)** is perhaps the single most important concept in finance. It states that **a rupee (or dollar) today is worth more than a rupee tomorrow**, because:

1. **Opportunity Cost:** Money available now can be invested to earn a return.
2. **Inflation:** Purchasing power erodes over time.
3. **Risk:** Future cash flows are uncertain.

2.2 Future Value (FV)

The **Future Value** is the amount a current sum grows to over time at a given interest rate.

Single Cash Flow:

$$FV = PV \times (1 + r)^n$$

Where:

- PV = Present Value
- r = Interest rate per period
- n = Number of periods

Example: ₹10,000 invested at 8% per annum for 5 years:

$$FV = 10,000 \times (1 + 0.08)^5$$

$$FV = 10,000 \times 1.4693$$

$$FV = ₹14,693.28$$

With Continuous Compounding:

$$FV = PV \times e^{(r \times t)}$$

2.3 Present Value (PV)

Present Value is the current worth of a future sum of money discounted at a specific rate.

$$PV = FV / (1 + r)^n$$

Example: What is the present value of ₹50,000 to be received in 7 years, with a discount rate of 10%?

$$PV = 50,000 / (1 + 0.10)^7$$

$$PV = 50,000 / 1.9487$$

$$PV = ₹25,658.19$$

2.4 Annuities

An **annuity** is a series of equal cash flows at regular intervals.

Present Value of Annuity:

$$PV_{\text{annuity}} = PMT \times [1 - (1 + r)^{-n}] / r$$

Future Value of Annuity:

$$FV_{\text{annuity}} = PMT \times [(1 + r)^n - 1] / r$$

Perpetuity (infinite annuity):

$$PV_{\text{perpetuity}} = PMT / r$$

Growing Perpetuity (Gordon Growth Model):

$$PV = PMT / (r - g)$$

Where g is the constant growth rate in cash flows.

2.5 Net Present Value (NPV) and Internal Rate of Return (IRR)

NPV is used to evaluate investment projects:

$$NPV = \sum [CF_t / (1 + r)^t] - \text{Initial Investment}$$

Decision Rule:

- $NPV > 0 \rightarrow$ Accept the project (value-creating)
- $NPV < 0 \rightarrow$ Reject the project (value-destroying)
- $NPV = 0 \rightarrow$ Break-even

IRR is the discount rate at which $NPV = 0$. It represents the project's expected rate of return:

- If $IRR > WACC$ (hurdle rate) $\rightarrow$ Accept
- If $IRR < WACC \rightarrow$ Reject

IRR Limitations:

- Multiple IRRs can exist for non-conventional cash flows
 - Ignores project scale (can favor smaller high-return projects over larger value-creating ones)
 - Assumes reinvestment at the IRR itself (unrealistic)
-

3. Equity Markets and Stock Valuation

3.1 Understanding Equity

Equity represents ownership in a company. Shareholders are residual claimants — they receive what is left after all other obligations (debt, taxes, preferred dividends) are paid.

Types of Shares:

- **Common Stock (Ordinary Shares):** Voting rights, residual claims on earnings
- **Preferred Stock:** Fixed dividend, no voting rights, priority over common in liquidation
- **American Depositary Receipts (ADRs):** Foreign company shares traded on US exchanges
- **Global Depositary Receipts (GDRs):** Similar, traded on international exchanges

3.2 Stock Exchanges

Exchange	Country	Founded	Notable Index
New York Stock Exchange (NYSE)	USA	1792	Dow Jones Industrial Average
NASDAQ	USA	1971	NASDAQ Composite
London Stock Exchange (LSE)	UK	1801	FTSE 100
Tokyo Stock Exchange (TSE)	Japan	1878	Nikkei 225
National Stock Exchange (NSE)	India	1992	NIFTY 50
Bombay Stock Exchange (BSE)	India	1875	SENSEX
Shanghai Stock Exchange	China	1990	SSE Composite
Hong Kong Exchange (HKEX)	Hong Kong	1891	Hang Seng Index

3.3 Stock Valuation Models

Dividend Discount Model (DDM)

The value of a stock equals the present value of all future dividends.

Gordon Growth Model (Constant Growth DDM):

$$P_0 = D_1 / (r - g)$$

Where:

- P_0 = Current stock price
- D_1 = Expected dividend next year
- r = Required rate of return
- g = Constant dividend growth rate

Multi-Stage DDM: Used when dividends grow at different rates during different phases.

Discounted Cash Flow (DCF) Valuation

$$\text{Intrinsic Value} = \sum [FCF_t / (1 + WACC)^t] + \text{Terminal Value} / (1 + WACC)^n$$

Terminal Value Calculation:

$$TV = FCF_n \times (1 + g) / (WACC - g)$$
$$TV = EBITDA_n \times \text{Exit Multiple}$$

[Gordon Growth Method]

[Exit Multiple Method]

Relative Valuation (Multiples)

Multiple	Formula	Typical Use
P/E Ratio	Price / Earnings per Share	General equity valuation
P/B Ratio	Price / Book Value per Share	Banks, asset-heavy companies
EV/EBITDA	Enterprise Value / EBITDA	Cross-industry comparison
EV/Sales	Enterprise Value / Revenue	High-growth, pre-profit companies
P/FCF	Price / Free Cash Flow	Cash-generative businesses
PEG Ratio	P/E / EPS Growth Rate	Growth-adjusted valuation

3.4 Market Capitalization Categories

Category	Market Cap (USD)	Examples
Mega Cap	> \$200 billion	Apple, Microsoft, Aramco
Large Cap	\$10B - \$200B	IBM, Ford, Pfizer
Mid Cap	\$2B - \$10B	Varied across sectors
Small Cap	\$300M - \$2B	Regional companies
Micro Cap	\$50M - \$300M	Early-stage companies
Nano Cap	< \$50M	Penny stocks

3.5 Technical Analysis vs. Fundamental Analysis

Fundamental Analysis:

- Examines financial statements, industry dynamics, macroeconomic conditions
- Estimates intrinsic value
- Time horizon: Long-term (months to years)
- Practitioners: Warren Buffett, Benjamin Graham (Value Investing)

Technical Analysis:

- Studies price charts, patterns, and trading volume
- Identifies trends and momentum
- Key tools: Moving Averages, RSI, MACD, Bollinger Bands, Fibonacci Retracements
- Time horizon: Short to medium-term

Common Technical Patterns:

- Head and Shoulders (bearish reversal)
 - Double Bottom (bullish reversal)
 - Cup and Handle (bullish continuation)
 - Descending Triangle (bearish continuation)
-

4. Fixed Income and Bond Markets

4.1 What is a Bond?

A **bond** is a debt instrument where the issuer borrows money from investors and agrees to pay:

1. **Coupon payments** — periodic interest payments (semi-annual in the US, annual in India/Europe)
2. **Face value (par value)** — typically ₹1,000 or \$1,000, repaid at maturity

Key Bond Terminology:

Term	Definition
Face Value / Par Value	Principal amount returned at maturity
Coupon Rate	Annual interest rate as % of face value
Coupon Payment	$\text{Coupon Rate} \times \text{Face Value (periodic)}$
Maturity Date	Date when principal is returned
Yield to Maturity (YTM)	Total return if held to maturity
Current Yield	$\text{Annual Coupon} / \text{Current Price}$
Duration	Weighted average time to receive cash flows (sensitivity to rates)
Convexity	Curvature of price-yield relationship

4.2 Types of Bonds

By Issuer:

- **Government Bonds:** G-Secs (India), Treasuries (USA), Gilts (UK)
- **Municipal Bonds:** Issued by state/local governments
- **Corporate Bonds:** Investment-grade vs. High-yield (Junk bonds)
- **Supranational Bonds:** World Bank, IMF, ADB issued bonds

By Structure:

- **Zero-Coupon Bonds:** Issued at discount, no periodic payments
- **Floating Rate Notes (FRNs):** Coupon tied to a benchmark (LIBOR, SOFR, MIBOR)
- **Callable Bonds:** Issuer can redeem before maturity
- **Puttable Bonds:** Investor can demand early repayment
- **Convertible Bonds:** Can be converted into equity at a specified price

By Maturity:

- **Bills / T-Bills:** < 1 year
- **Notes:** 1-10 years
- **Bonds:** > 10 years (30-year bonds are common)
- **Perpetuals / Consols:** No maturity, pay coupons forever

4.3 Bond Pricing

The price of a bond is the present value of all future cash flows:

$$\text{Bond Price} = \sum [C / (1 + r)^t] + [F / (1 + r)^n]$$

Where:

- C = Periodic coupon payment
- F = Face value
- r = Required yield (YTM) per period
- n = Total number of periods

Bond Price and Yield Relationship (Inverse):

- If YTM > Coupon Rate → Bond trades at a **Discount** (Price < Par)
- If YTM = Coupon Rate → Bond trades at **Par** (Price = Par)
- If YTM < Coupon Rate → Bond trades at a **Premium** (Price > Par)

4.4 Duration and Interest Rate Risk

Macaulay Duration measures the weighted average time (in years) to receive cash flows. It quantifies interest rate sensitivity.

$$\text{Macaulay Duration} = \sum [t \times \text{PV}(\text{CF}_t)] / \text{Bond Price}$$

Modified Duration:

$$\text{Modified Duration} = \text{Macaulay Duration} / (1 + \text{YTM}/m)$$

Price Change Approximation:

$$\Delta P/P \approx -\text{Modified Duration} \times \Delta y$$

Example: A bond with Modified Duration = 7.5 will fall approximately 7.5% in price if yields rise by 1%.

Convexity corrects for the non-linear price-yield relationship:

$$\Delta P/P \approx -D^* \times \Delta y + (1/2) \times \text{Convexity} \times (\Delta y)^2$$

Higher convexity is beneficial to bondholders (upside greater, downside smaller).

4.5 The Yield Curve

The **yield curve** plots bond yields against their maturities. It signals economic expectations:

Yield Curve Shape	Description	Economic Signal
Normal (Upward Sloping)	Short rates < Long rates	Healthy economic growth expected
Inverted (Downward Sloping)	Short rates > Long rates	Recession warning (historically reliable)
Flat	Short rates $\approx$ Long rates	Economic transition, uncertainty
Humped	Medium rates highest	Complex expectations

Credit Spread = Corporate Bond Yield – Risk-Free (Government) Bond Yield

- Wider spreads signal higher perceived credit risk

4.6 Credit Ratings

Moody's	S&P / Fitch	Category
Aaa	AAA	Highest quality
Aa1, Aa2, Aa3	AA+, AA, AA−	High quality
A1, A2, A3	A+, A, A−	Upper medium grade
Baa1, Baa2, Baa3	BBB+, BBB, BBB−	Lower medium grade (Investment Grade boundary)
Ba1, Ba2	BB+, BB	Speculative / High Yield (Junk) begins
B1-B3	B+, B, B−	Highly speculative
Caa-C	CCC-C	Near default / In default
D	D	Default

5. Derivatives: Options, Futures, and Swaps

5.1 What are Derivatives?

A **derivative** is a financial contract whose value is derived from an underlying asset. Underlying assets include:

- Equities (individual stocks, indices)
- Commodities (gold, oil, wheat)
- Currencies (USD/INR, EUR/USD)
- Interest rates
- Credit events

Purposes of Derivatives:

1. **Hedging:** Reduce risk exposure
2. **Speculation:** Profit from price movements with leverage
3. **Arbitrage:** Exploit price discrepancies

5.2 Forward Contracts

A **forward contract** is a customized agreement to buy/sell an asset at a specified price on a future date.

- Traded **OTC** (over-the-counter), not on exchanges
- Customizable in size, date, and delivery location
- Carries **counterparty risk**
- Common in currencies (FX Forwards) and commodities

Forward Price Formula:

$$F_0 = S_0 \times e^{(r-q)T}$$

Where:

- S_0 = Current spot price
- r = Risk-free rate
- q = Dividend yield / convenience yield
- T = Time to expiry (in years)

5.3 Futures Contracts

Futures are standardized, exchange-traded forward contracts with daily mark-to-market settlement.

Feature	Forward	Futures
Trading venue	OTC	Exchange (NSE, CME, NYMEX)
Standardization	Customizable	Standardized (size, date)
Counterparty Risk	Yes	No (Clearinghouse steps in)
Settlement	End of contract	Daily (MTM)
Margin	Not required	Required (Initial + Maintenance)

Key Futures Markets:

- **Equity Index Futures:** NIFTY50, S&P 500, DAX
- **Commodity Futures:** Crude Oil (WTI, Brent), Gold, Silver, Natural Gas
- **Currency Futures:** USD/INR, EUR/USD
- **Interest Rate Futures:** 10-Year T-Bond Futures

5.4 Options

An **option** gives the buyer the **right, but not the obligation**, to buy or sell an asset at a predetermined price (strike price) on or before a specific date.

Types of Options:

- **Call Option:** Right to **BUY** the underlying
- **Put Option:** Right to **SELL** the underlying

Option Styles:

- **American:** Can be exercised any time before expiry
- **European:** Can only be exercised at expiry

Option Profit/Loss:

Position	When Profitable	Max Profit	Max Loss
Long Call	Stock rises above Strike + Premium	Unlimited	Premium paid
Short Call	Stock stays below Strike	Premium received	Unlimited
Long Put	Stock falls below Strike – Premium	Strike – Premium	Premium paid
Short Put	Stock stays above Strike	Premium received	Strike – Premium

5.5 The Black-Scholes Model

Developed in 1973 by Fischer Black, Myron Scholes, and Robert Merton, the **Black-Scholes model** prices European options.

Call Option Price:

$$C = S_0 N(d_1) - Ke^{(-rT)} N(d_2)$$

Put Option Price:

$$P = Ke^{(-rT)} N(-d_2) - S_0 N(-d_1)$$

Where:

$$d_1 = [\ln(S_0/K) + (r + \sigma^2/2)T] / (\sigma\sqrt{T})$$
$$d_2 = d_1 - \sigma\sqrt{T}$$

- S_0 = Current stock price
- K = Strike price
- T = Time to expiry
- r = Risk-free rate
- σ = Volatility (standard deviation of returns)
- $N(\cdot)$ = Cumulative normal distribution function

5.6 The Greeks

The **Greeks** measure an option's sensitivity to various parameters:

Greek	Symbol	Measures	Typical Values
Delta	Δ	Price change per \$1 move in underlying	Calls: 0 to 1, Puts: -1 to 0
Gamma	Γ	Rate of change of Delta	Highest ATM, near expiry
Theta	Θ	Time decay (value lost per day)	Negative for long options
Vega	v	Sensitivity to volatility change	Positive for long options
Rho	ρ	Sensitivity to interest rate change	Positive for calls, negative for puts

5.7 Common Options Strategies

Strategy	Construction	View	Max Profit	Max Loss
Covered Call	Long Stock + Short Call	Neutral to mildly bullish	Strike - Cost + Premium	Stock goes to 0
Protective Put	Long Stock + Long Put	Bullish but hedged	Unlimited	Limited by put
Bull Call Spread	Long lower strike call + Short higher strike call	Moderately bullish	Spread width - Premium	Net premium
Bear Put Spread	Long higher strike put + Short lower strike put	Moderately bearish	Spread - Net Premium	Net premium
Straddle	Long Call + Long Put (same strike)	High volatility expected	Unlimited	Total premium
Iron Condor	Bull put spread + Bear call spread	Low volatility expected	Net premium	Spread - premium

5.8 Swaps

A **swap** is an agreement between two parties to exchange cash flows over time.

Interest Rate Swap (IRS):

- Party A pays a **fixed rate**, receives a **floating rate** (e.g., LIBOR/SOFR)
- Party B pays **floating**, receives **fixed**
- Used to manage interest rate exposure

Currency Swap:

- Exchange principal and interest payments in different currencies
- Used by multinationals to hedge FX exposure

Credit Default Swap (CDS):

- Buyer pays periodic premium; seller compensates if the reference entity defaults
 - Functions like "insurance" on credit risk
 - Played a major role in the 2008 financial crisis
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6. Portfolio Theory and Asset Allocation

6.1 Modern Portfolio Theory (MPT)

Developed by Harry Markowitz in 1952, **Modern Portfolio Theory** formalizes how investors can construct optimal portfolios by balancing expected return and risk.

Key Insight: Through diversification, investors can reduce **unsystematic risk** (company-specific) without necessarily reducing expected returns.

Expected Return of a Portfolio:

$$E(R_p) = \sum w_i \times E(R_i)$$

Portfolio Variance (two assets):

$$\begin{aligned}\sigma^2_p &= w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \text{Cov}(R_1, R_2) \\ \sigma^2_p &= w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{1,2} \sigma_1 \sigma_2\end{aligned}$$

Where $\rho_{1,2}$ is the correlation between assets 1 and 2.

Diversification Benefit:

- When $\rho = +1$: No diversification benefit
- When $\rho = 0$: Partial diversification benefit
- When $\rho = -1$: Maximum diversification (risk eliminated)

6.2 The Efficient Frontier

The **Efficient Frontier** is the set of portfolios that offer the highest expected return for a given level of risk.

- Portfolios **on** the frontier: Optimal (no more return without more risk)
- Portfolios **below** the frontier: Inefficient (dominated by frontier portfolios)
- Portfolios **above** the frontier: Impossible to achieve

The **Minimum Variance Portfolio (MVP)** is the point on the frontier with the lowest possible volatility.

6.3 Capital Market Line (CML) and CAPM

When a **risk-free asset** is introduced, the optimal portfolio combines the risk-free asset with the **Market Portfolio** (all risky assets in proportion to their market cap).

Capital Market Line:

$$E(R_p) = R_f + [(E(R_m) - R_f) / \sigma_m] \times \sigma_p$$

Capital Asset Pricing Model (CAPM):

$$E(R_i) = R_f + \beta_i \times [E(R_m) - R_f]$$

Where:

- R_f = Risk-free rate
- β_i = Beta of asset i (systematic risk measure)
- $E(R_m)$ = Expected market return
- $[E(R_m) - R_f]$ = Equity Risk Premium (ERP)

Beta Interpretation:

- $\beta = 1$: Asset moves in line with market
- $\beta > 1$: Asset is more volatile than market (aggressive)
- $\beta < 1$: Asset is less volatile (defensive)
- $\beta < 0$: Asset moves opposite to market (rare, e.g., inverse ETFs)

6.4 Alpha

Alpha (α) is the excess return of an investment above what CAPM predicts:

$$\alpha = \text{Actual Return} - \text{CAPM Expected Return}$$
$$\alpha = R_i - [R_f + \beta(R_m - R_f)]$$

Positive alpha indicates outperformance; it's the "holy grail" of active management.

6.5 Factor Models

Fama-French Three-Factor Model (1992): Adds two factors to CAPM:

$$E(R_i) - R_f = \beta_1(R_m - R_f) + \beta_2(SMB) + \beta_3(HML)$$

- **SMB (Small Minus Big):** Small-cap stocks outperform large-cap
- **HML (High Minus Low):** Value stocks (high B/P) outperform growth stocks

Carhart Four-Factor Model: Adds **Momentum (MOM)** factor.

Fama-French Five-Factor Model (2015): Adds **Profitability (RMW)** and **Investment (CMA)** factors.

6.6 Asset Allocation Strategies

Strategic Asset Allocation (SAA): Long-term target weights (e.g., 60% equity, 40% bonds)

Tactical Asset Allocation (TAA): Short-term deviations from SAA to exploit market opportunities

Dynamic Asset Allocation: Continuously adjusts weights based on market conditions

Risk Parity: Allocates based on equal risk contribution from each asset class, not equal capital

Example SAA by Investor Profile:

Profile	Equity	Fixed Income	Cash/Alternatives
Aggressive (Young, Long Horizon)	80%	15%	5%
Moderate (Mid-Career)	60%	30%	10%
Conservative (Near Retirement)	35%	55%	10%
Capital Preservation	15%	70%	15%

7. Corporate Finance and Capital Structure

7.1 The Goal of Corporate Finance

The primary objective of financial management is to **maximize shareholder wealth**, which translates to maximizing the firm's stock price or intrinsic value. This involves three key decisions:

1. **Investment Decision (Capital Budgeting):** Which projects to invest in?
2. **Financing Decision (Capital Structure):** How to fund those investments?
3. **Dividend Decision:** How much of earnings to return to shareholders?

7.2 Capital Structure

Capital structure refers to the mix of debt and equity a company uses to finance its assets.

$$\begin{aligned}\text{Total Assets} &= \text{Total Debt} + \text{Total Equity} \\ \text{Enterprise Value} &= \text{Market Cap} + \text{Net Debt}\end{aligned}$$

Leverage Ratios:

- Debt-to-Equity (D/E) = Total Debt / Total Equity
- Debt-to-Capital = Total Debt / (Debt + Equity)
- Interest Coverage = EBIT / Interest Expense

7.3 Modigliani-Miller Theorems

Without taxes (1958): Capital structure is irrelevant — firm value is independent of financing. A firm cannot change its total value by splitting cash flows differently between debt and equity.

With taxes (1963): Debt is beneficial because **interest payments are tax-deductible**, creating a "tax shield":

$$\begin{aligned}\text{Value of Levered Firm} &= \text{Value of Unlevered Firm} + \text{PV}(\text{Tax Shield}) \\ \text{Value of Levered Firm} &= V_U + t \times D\end{aligned}$$

Where t = corporate tax rate, D = debt outstanding.

Implication: Firms should use as much debt as possible to maximize tax shields... but this ignores financial distress costs.

7.4 Trade-off Theory

The **Trade-off Theory** finds the optimal capital structure where:

$$\text{Optimal Debt} = \text{Point where Marginal Benefit of Tax Shield} = \text{Marginal Cost of Financial Distress}$$

Costs of Financial Distress:

- Direct costs: Legal/bankruptcy fees
- Indirect costs: Loss of customers, suppliers, employees; management distraction

7.5 Weighted Average Cost of Capital (WACC)

WACC is the blended cost of capital from all sources, used as the discount rate for firm-level DCF valuations:

$$WACC = (E/V) \times Re + (D/V) \times Rd \times (1 - t)$$

Where:

- E = Market value of equity
- D = Market value of debt
- V = $E + D$ (total firm value)
- Re = Cost of equity
- Rd = Cost of debt (pre-tax)
- t = Corporate tax rate

Cost of Equity (using CAPM):

$$Re = Rf + \beta \times (Rm - Rf)$$

Example WACC Calculation:

- Equity = ₹600 Cr, Debt = ₹400 Cr, Total = ₹1,000 Cr
- Cost of Equity = 14%, Cost of Debt = 8%, Tax Rate = 25%

$$\begin{aligned} WACC &= (600/1000) \times 14\% + (400/1000) \times 8\% \times (1 - 0.25) \\ WACC &= 0.60 \times 14\% + 0.40 \times 6\% \\ WACC &= 8.4\% + 2.4\% = 10.8\% \end{aligned}$$

7.6 Dividend Policy

Gordon Growth Model (revisited): Dividends signal management's confidence in future earnings.

Dividend Policy Types:

- **Stable Dividend Policy:** Constant dividends; most common, preferred by income investors
- **Residual Dividend Policy:** Pay dividends only after all positive-NPV investments are funded
- **Constant Payout Ratio:** Fixed % of earnings each year
- **Zero Dividend / Share Buybacks:** Reinvest all earnings; prefer buybacks for tax efficiency

Modigliani-Miller Dividend Irrelevance: In perfect markets, dividend policy doesn't affect firm value — investors can create "homemade dividends" by selling shares.

In practice: Dividends and buybacks do matter due to taxes, signaling effects, and investor preferences.

7.7 Mergers and Acquisitions (M&A)

Types of M&A:

- **Horizontal Merger:** Same industry (e.g., two banks merging)
- **Vertical Merger:** Supply chain integration (e.g., manufacturer acquires supplier)
- **Conglomerate Merger:** Unrelated businesses
- **Acquisition:** One company buys another (friendly or hostile)
- **Leveraged Buyout (LBO):** Private equity buys company using high leverage

Valuation in M&A:

- Comparable Company Analysis (Comps)
- Precedent Transaction Analysis
- DCF Analysis
- Accretion/Dilution Analysis

Synergies = Additional value created by combining firms:

$$\text{Deal Value} = \text{Standalone Value of Target} + \text{PV}(\text{Synergies}) - \text{Premium Paid}$$

8. Financial Statement Analysis

8.1 The Three Core Financial Statements

Every public company must publish three core financial statements:

1. **Income Statement (P&L):** Shows revenues and expenses over a period
2. **Balance Sheet:** Shows assets, liabilities, and equity at a point in time
3. **Cash Flow Statement:** Shows cash inflows and outflows over a period

8.2 Income Statement Structure

Revenue (Net Sales)
- Cost of Goods Sold (COGS)
= Gross Profit
- Operating Expenses (SG&A, R&D, Depreciation)
= EBIT (Earnings Before Interest and Taxes)
- Interest Expense
= EBT (Earnings Before Tax)
- Income Tax Expense
= Net Income
- Preferred Dividends
= Earnings Available to Common Shareholders
÷ Shares Outstanding
= EPS (Earnings Per Share)

8.3 Balance Sheet Structure

Assets = Liabilities + Shareholders' Equity

Assets	Liabilities & Equity
Current Assets	Current Liabilities
Cash & Equivalents	Accounts Payable
Accounts Receivable	Short-term Debt
Inventory	Accrued Expenses
Prepaid Expenses	Deferred Revenue
Non-Current Assets	Long-term Liabilities
Property, Plant & Equipment	Long-term Debt
Intangible Assets (Goodwill)	Deferred Tax Liability
Long-term Investments	Shareholders' Equity
	Common Stock + Retained Earnings

8.4 Cash Flow Statement

Three Sections:

1. **Operating Activities:** Cash from core business operations (Net Income adjusted for non-cash items and working capital changes)
2. **Investing Activities:** Capital expenditures, acquisitions, asset sales
3. **Financing Activities:** Debt issuance/repayment, equity issuance, dividends paid

Free Cash Flow (FCF):

$$\text{FCF} = \text{Operating Cash Flow} - \text{Capital Expenditures (CapEx)}$$

Unlevered FCF (for DCF):

$$\text{UFCF} = \text{EBIT} \times (1 - \text{Tax Rate}) + \text{D\&A} - \Delta \text{Working Capital} - \text{CapEx}$$

8.5 Profitability Ratios

Ratio	Formula	What It Measures
Gross Margin	Gross Profit / Revenue	Efficiency of production/service delivery
Operating Margin (EBIT Margin)	EBIT / Revenue	Core operational efficiency
Net Profit Margin	Net Income / Revenue	Overall bottom-line profitability
Return on Assets (ROA)	Net Income / Total Assets	How well assets generate profit
Return on Equity (ROE)	Net Income / Shareholders' Equity	Return on equity capital
Return on Invested Capital (ROIC)	NOPAT / Invested Capital	Quality of capital allocation
EBITDA Margin	EBITDA / Revenue	Cash profitability (pre capex)

8.6 DuPont Analysis

DuPont breaks ROE into component drivers:

3-Factor DuPont:

$$\begin{aligned} \text{ROE} &= \text{Net Profit Margin} \times \text{Asset Turnover} \times \text{Financial Leverage} \\ \text{ROE} &= (\text{NI/Sales}) \times (\text{Sales/Assets}) \times (\text{Assets/Equity}) \end{aligned}$$

5-Factor DuPont:

$$\text{ROE} = \text{Tax Burden} \times \text{Interest Burden} \times \text{EBIT Margin} \times \text{Asset Turnover} \times \text{Leverage}$$

This reveals whether ROE improvement comes from higher profitability, better asset efficiency, or increased leverage.

9. Risk Management

9.1 Types of Financial Risk

Risk Type	Description	Examples
Market Risk	Loss from adverse market movements	Stock decline, interest rate rise
Credit Risk	Counterparty defaults on obligations	Loan default, bond issuer bankruptcy
Liquidity Risk	Cannot sell asset quickly at fair value	Illiquid bonds, real estate
Operational Risk	Losses from processes, systems, people	Fraud, system failure, human error
Currency Risk (FX)	Losses from exchange rate movements	USD/INR appreciation hurting exports
Interest Rate Risk	Changes in rates affect portfolio value	Rising rates reducing bond prices
Inflation Risk	Real returns eroded by inflation	Fixed-income investments
Political / Regulatory Risk	Government actions affect investments	Tax changes, nationalization
Model Risk	Errors in models used for pricing/risk	Incorrect Black-Scholes assumptions

9.2 Value at Risk (VaR)

VaR answers: "What is the maximum loss we can expect with X% confidence over Y days?"

Definition: At 95% confidence over 1 day, a VaR of ₹10 Cr means there is a 5% chance of losing more than ₹10 Cr in a single day.

Three Approaches:

1. **Historical Simulation:** Use actual historical returns to compute loss distribution
2. **Parametric (Delta-Normal) VaR:** Assumes normal distribution

$$\text{VaR} = \text{Portfolio Value} \times z\text{-score} \times \sigma \times \sqrt{T}$$

($z = 1.645$ for 95% confidence, 2.326 for 99%)

3. **Monte Carlo Simulation:** Simulate thousands of random scenarios

Limitations of VaR:

- Doesn't capture "tail risk" beyond the confidence level
- Assumes historical patterns continue
- Not sub-additive in all cases (violates diversification logic)

9.3 Expected Shortfall (CVaR)

Expected Shortfall (ES) / Conditional VaR (CVaR) measures the average loss *beyond* the VaR threshold. It is considered superior to VaR for tail risk management.

$$\text{CVaR}_{95\%} = \text{Average of all losses exceeding the 95\% VaR}$$

9.4 Hedging Strategies

Natural Hedging: Matching revenues and costs in the same currency/rate environment (e.g., an Indian exporter borrowing in USD).

Derivative Hedging:

- **Futures/Forwards:** Lock in a future price
- **Options:** Asymmetric protection (like insurance)
- **Swaps:** Convert variable-rate to fixed-rate exposure (or vice versa)

Hedge Ratio (for futures):

$$\text{Optimal Hedge Ratio} = \rho \times (\sigma_{\text{spot}} / \sigma_{\text{futures}})$$

Beta Hedging (equity portfolio):

$$\text{Number of Contracts} = (\beta_{\text{target}} - \beta_{\text{portfolio}}) / \beta_{\text{futures}} \times (\text{Portfolio Value} / \text{Futures Value})$$

10. Macroeconomics and Monetary Policy

10.1 Key Macroeconomic Indicators

Indicator	Measures	Data Source (India)
GDP Growth Rate	Economic output growth	MOSPI
CPI Inflation	Consumer price level changes	MoSPI / RBI
WPI (Wholesale Price Index)	Wholesale price changes	DPIIT
Repo Rate	RBI's key lending rate	RBI
Unemployment Rate	Labour market slack	CMIE / PLFS
Current Account Deficit (CAD)	Trade + income balance	RBI
Fiscal Deficit	Government spending - Revenue	Budget Documents
Foreign Exchange Reserves	RBI's FX holdings	RBI Weekly
Index of Industrial Production (IIP)	Manufacturing activity	MOSPI
PMI (Purchasing Managers' Index)	Business activity sentiment	S&P Global

10.2 The Business Cycle

The economy moves through four phases:

1. **Expansion:** Rising GDP, falling unemployment, increasing corporate profits, stock markets typically rise
2. **Peak:** Maximum economic output, inflation risks rise, central bank tightens
3. **Contraction/Recession:** Falling GDP (two consecutive quarters), rising unemployment, credit tightens
4. **Trough:** Minimum output, central banks ease, recovery begins

Cyclical vs. Defensive Sectors:

Cyclical (Benefit in Expansion)	Defensive (Hold in Contraction)
Consumer Discretionary	Consumer Staples
Industrials	Healthcare
Materials	Utilities
Technology	Pharmaceuticals
Financials	FMCG

10.3 Monetary Policy

Central banks use monetary policy to achieve price stability and economic growth.

RBI Policy Tools:

- **Repo Rate:** Rate at which RBI lends to commercial banks (key policy rate)
- **Reverse Repo Rate:** Rate at which RBI borrows from commercial banks
- **Cash Reserve Ratio (CRR):** Mandatory % of deposits kept with RBI
- **Statutory Liquidity Ratio (SLR):** % of deposits in liquid assets (G-Secs, cash)
- **Open Market Operations (OMO):** Buying/selling G-Secs to inject/absorb liquidity
- **Marginal Standing Facility (MSF):** Emergency borrowing window

Monetary Policy Stances:

- **Accommodative:** Low rates, support growth (used during downturns)
- **Neutral:** Neither tightening nor easing
- **Hawkish:** Tightening bias, combat inflation
- **Withdrawal of Accommodation:** Gradual removal of stimulus

10.4 Inflation

Inflation is the rate at which the general price level increases over time.

Causes:

- **Demand-Pull:** Economy overheats, too much money chasing goods
- **Cost-Push:** Rising input costs (oil prices, wages) push prices up
- **Structural:** Supply bottlenecks, poor productivity

Measures:

- **CPI (Consumer Price Index):** Most common; tracks retail prices of a basket of goods
- **Core CPI:** Excludes food and fuel (volatile items)
- **WPI:** Tracks wholesale/producer price changes
- **PCE Deflator:** US Federal Reserve's preferred measure

Fisher Equation:

$$\text{Nominal Interest Rate} \approx \text{Real Interest Rate} + \text{Inflation Rate}$$

$$(1 + \text{nominal}) = (1 + \text{real}) \times (1 + \text{inflation})$$

Real Returns:

$$\text{Real Return} = [(1 + \text{Nominal Return}) / (1 + \text{Inflation Rate})] - 1$$

10.5 The Fiscal Policy

Fiscal policy involves government spending and taxation decisions.

Expansionary Fiscal Policy:

- Increase government spending
- Decrease taxes
- → Stimulates aggregate demand
- → Often leads to higher fiscal deficit

Contractionary Fiscal Policy:

- Decrease spending
- Increase taxes
- → Reduces inflationary pressures

Fiscal Deficit:

$$\text{Fiscal Deficit} = \text{Total Expenditure} - \text{Total Receipts (excluding borrowings)}$$

India's fiscal deficit target is typically ~4.5-5.1% of GDP (Fiscal Consolidation Road Map).

Debt-to-GDP Ratio: Measures government debt sustainability. IMF suggests < 60% for emerging markets is advisable.

11. Alternative Investments

11.1 What are Alternative Investments?

Alternative investments are asset classes outside traditional stocks, bonds, and cash. They are typically:

- Less liquid
- Higher returning over long horizons
- Less correlated with public markets
- Suitable for institutional investors and HNIs (High Net Worth Individuals)

11.2 Private Equity

Private equity involves investing in companies not listed on public exchanges.

Types:

- **Venture Capital (VC):** Early-stage startups (Seed, Series A/B/C)
- **Growth Equity:** Established companies seeking expansion capital
- **Leveraged Buyouts (LBO):** Acquiring mature companies using debt
- **Distressed/Special Situations:** Investing in financially troubled companies

PE Returns Metrics:

- **IRR (Internal Rate of Return):** Annualized return on invested capital
- **MOIC (Multiple on Invested Capital):** Total value / invested capital
- **DPI (Distributions to Paid-In Capital):** Cash returned / invested capital
- **RVPI (Residual Value to Paid-In):** Remaining unrealized value / invested capital
- **TVPI = DPI + RVPI**

LBO Model Logic:

Returns = f(Entry Multiple, Exit Multiple, Revenue Growth, Margin Expansion, Leverage)

11.3 Real Estate

Real Estate Investment Types:

- **Direct Ownership:** Residential, commercial, industrial properties
- **REITs (Real Estate Investment Trusts):** Liquid, exchange-listed exposure to real estate
- **Real Estate Debt:** Mortgage lending, mezzanine financing
- **Infrastructure:** Roads, airports, power plants, data centers

Key RE Metrics:

- **Cap Rate (Capitalization Rate):** $\text{NOI} / \text{Property Value}$ (analogous to earnings yield)
- **NOI (Net Operating Income):** Revenue - Operating Expenses (before debt service)
- **Cash-on-Cash Return:** Annual Cash Flow / Cash Invested
- **LTV (Loan-to-Value):** Mortgage / Property Value

11.4 Hedge Funds

Hedge funds are lightly regulated investment vehicles that use sophisticated strategies to generate returns.

Common Strategies:

Strategy	Description	Risk/Return
Long/Short Equity	Long undervalued, short overvalued stocks	Moderate
Global Macro	Bets on macroeconomic trends (currencies, rates, commodities)	High
Statistical Arbitrage	Quantitative pairs trading	Low-Moderate
Event Driven	Mergers, bankruptcies, spin-offs	Moderate
Distressed Debt	Buys bonds/loans of struggling companies	High
Quantitative / CTA	Algorithmic, trend-following strategies	Varies

Fee Structure: Typically "2 and 20" — 2% management fee + 20% performance fee above a hurdle rate.

11.5 Commodities

Commodity Categories:

- **Energy:** Crude Oil (WTI, Brent), Natural Gas, Coal
- **Precious Metals:** Gold, Silver, Platinum
- **Base Metals:** Copper, Aluminum, Zinc, Nickel
- **Agricultural:** Wheat, Corn, Soybeans, Sugar, Cotton
- **Soft Commodities:** Coffee, Cocoa, Orange Juice

Gold in Finance:

- Often seen as a **safe haven** during market stress
- **Inverse correlation** with USD (generally)
- **Store of value** against inflation
- Central bank reserves include significant gold holdings

12. FinTech and Digital Finance

12.1 The FinTech Revolution

FinTech (Financial Technology) refers to companies and technologies that improve financial services. Key areas include:

- **Digital Payments:** UPI (India), PayPal, Stripe, Square
- **Neobanks:** Digital-only banks (Revolut, N26, Fi Money, Jupiter)
- **InsurTech:** AI-driven insurance (Acko, Digit Insurance)
- **WealthTech:** Robo-advisors, digital wealth platforms (Zerodha, Groww, Smallcase)
- **LendingTech:** P2P lending, BNPL (Buy Now Pay Later)
- **RegTech:** Technology for regulatory compliance

12.2 Cryptocurrencies and Blockchain

Blockchain is a distributed ledger technology (DLT) where transactions are recorded in blocks chained cryptographically.

Key Concepts:

- **Proof of Work (PoW):** Miners solve computational puzzles (Bitcoin uses this)
- **Proof of Stake (PoS):** Validators stake crypto as collateral (Ethereum post-merge)
- **DeFi (Decentralized Finance):** Financial services without intermediaries
- **Smart Contracts:** Self-executing code on blockchain (Ethereum)
- **NFTs (Non-Fungible Tokens):** Unique digital assets on blockchain
- **Stablecoins:** Crypto pegged to a fiat currency (USDT, USDC, DAI)

Major Cryptocurrencies:

Crypto	Symbol	Market Cap (2024)	Use Case
Bitcoin	BTC	~\$1.3 Trillion	Digital gold, store of value
Ethereum	ETH	~\$400 Billion	Smart contracts, DeFi platform
Binance Coin	BNB	~\$90 Billion	Binance exchange utility
Solana	SOL	~\$80 Billion	High-speed blockchain
Ripple	XRP	~\$60 Billion	Cross-border payments

Regulatory Landscape (India):

- Crypto income taxed at 30% flat (Budget 2022)
- 1% TDS on crypto transactions
- RBI exploring a CBDC (**Central Bank Digital Currency**) — the Digital Rupee (e-₹)

12.3 Open Banking and APIs

Open Banking requires banks to share customer data (with consent) with third-party providers via APIs.

India's Ecosystem:

- **Account Aggregator (AA) Framework:** Consent-based financial data sharing
 - **OCEN (Open Credit Enablement Network):** Democratizes credit access for MSMEs
 - **UPI (Unified Payments Interface):** Real-time payment infrastructure processed by NPCI
-

13. Behavioral Finance

13.1 Why Markets Aren't Always Rational

Behavioral finance integrates psychology and economics to explain why investors often act irrationally. Key founding figures: Daniel Kahneman, Amos Tversky, Richard Thaler.

13.2 Common Cognitive Biases

Bias	Description	Example in Investing
Overconfidence	Investors overestimate their ability	Excessive trading, underestimating risk
Anchoring	Over-reliance on first piece of information	Fixating on a stock's 52-week high
Confirmation Bias	Seeking information that confirms existing beliefs	Only reading bullish analysis
Loss Aversion	Losses hurt ~2× more than gains feel good	Holding losers too long, selling winners too early
Herding	Following the crowd	Buying at market peaks during euphoria
Mental Accounting	Treating money differently based on source	Gambling with "house money" from market gains
Recency Bias	Overweighting recent events	Buying after a bull market thinking it will continue
Availability Heuristic	Judging probability by ease of recall	Overestimating plane crash risk after news coverage
Sunk Cost Fallacy	Continuing bad investment because of past costs	"I can't sell now, I'd lose money"
Disposition Effect	Selling winners too early, holding losers too long	Combination of loss aversion + mental accounting

13.3 Prospect Theory

Developed by Kahneman and Tversky (1979), **Prospect Theory** is an alternative to Expected Utility Theory:

1. **Reference Dependence:** Outcomes evaluated as gains/losses relative to a reference point
2. **Loss Aversion:** Losses are felt more powerfully than equivalent gains
3. **Diminishing Sensitivity:** Each additional gain/loss matters less as it grows
4. **Probability Weighting:** People overweight small probabilities, underweight large ones

Value Function: Concave for gains (risk averse), convex for losses (risk seeking in loss domain).

14. Key Financial Ratios Quick Reference

Profitability

Ratio	Formula	Good Level (General)
Gross Margin	Gross Profit / Revenue	Higher is better; varies by industry
Net Margin	Net Income / Revenue	Positive; > 10% is good
ROE	Net Income / Equity	> 15% for most industries
ROIC	NOPAT / Invested Capital	> WACC (value creation)
ROA	Net Income / Total Assets	> 5% generally

Liquidity

Ratio	Formula	Healthy Range
Current Ratio	Current Assets / Current Liabilities	1.5 – 3.0
Quick Ratio (Acid Test)	(Current Assets - Inventory) / Current Liabilities	> 1.0
Cash Ratio	Cash / Current Liabilities	> 0.5

Leverage / Solvency

Ratio	Formula	Note
Debt-to-Equity	Total Debt / Total Equity	< 2.0 generally
Debt-to-EBITDA	Total Debt / EBITDA	< 3.0x generally; > 5x is risky
Interest Coverage	EBIT / Interest Expense	> 3.0 (comfortable)
Asset Coverage	(Assets - Intangibles - CL) / Debt	> 1.5

Efficiency

Ratio	Formula	Note
Asset Turnover	Revenue / Total Assets	Higher = better asset utilization
Inventory Turnover	COGS / Average Inventory	Higher = faster inventory movement
Receivables Turnover	Revenue / Average A/R	Higher = faster collections
Days Sales Outstanding (DSO)	365 / Receivables Turnover	Lower = faster cash collection
Cash Conversion Cycle (CCC)	DIO + DSO - DPO	Lower is better

Valuation

Ratio	Formula	Note
P/E Ratio	Market Price / EPS	Compare to sector average
Forward P/E	Price / Next Year EPS	Forward-looking
PEG Ratio	P/E / EPS Growth Rate	< 1 may indicate undervaluation
P/B Ratio	Market Cap / Book Value	Banks: ~1-2 is typical
EV/EBITDA	Enterprise Value / EBITDA	8-15x typical for most sectors
Dividend Yield	Annual DPS / Stock Price	Higher = more income

Appendix: Key Finance Formulas Summary

TIME VALUE OF MONEY

Future Value:	$FV = PV \times (1 + r)^n$
Present Value:	$PV = FV / (1 + r)^n$
PV Annuity:	$PV = PMT \times [1 - (1+r)^{-n}] / r$
FV Annuity:	$FV = PMT \times [(1+r)^n - 1] / r$
Perpetuity:	$PV = PMT / r$
Growing Perpetuity:	$PV = PMT / (r - g)$

BONDS

Bond Price:	$P = \sum C / (1+r)^t + F / (1+r)^n$
Modified Duration:	$MD = \text{Macaulay Duration} / (1 + YTM/m)$
Price Change:	$\Delta P/P \approx -MD \times \Delta y$

EQUITY VALUATION

DDM (Gordon Growth):	$P = D_1 / (r - g)$
DCF:	$V = \sum FCF_t / (1+WACC)^t + TV / (1+WACC)^n$
Terminal Value:	$TV = FCF \times (1+g) / (WACC - g)$

PORTFOLIO THEORY

Portfolio Return:	$R_p = \sum w_i R_i$
Portfolio Variance:	$\sigma^2 p = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho \sigma_1 \sigma_2$
CAPM:	$E(R_i) = R_f + \beta \times (R_m - R_f)$
Alpha:	$\alpha = \text{Actual Return} - \text{CAPM Return}$
Sharpe Ratio:	$(R_p - R_f) / \sigma_p$

CORPORATE FINANCE

WACC:	$(E/V) \times R_e + (D/V) \times R_d \times (1-t)$
MM (with tax):	$VL = VU + t \times D$
NPV:	$\sum CF_t / (1+r)^t - \text{Investment}$
Free Cash Flow:	$FCF = \text{Operating CF} - \text{CapEx}$

OPTIONS (BLACK-SCHOLES)

Call:	$C = S_0 N(d_1) - Ke^{(-rT)} N(d_2)$
Put:	$P = Ke^{(-rT)} N(-d_2) - S_0 N(-d_1)$

$$d_1 = [\ln(S/K) + (r + \sigma^2/2)T] / (\sigma\sqrt{T})$$
$$d_2 = d_1 - \sigma\sqrt{T}$$

RISK

Parametric VaR:	Portfolio × z × σ × √T
Sharpe Ratio:	(Return - Rf) / StdDev
Sortino Ratio:	(Return - Rf) / Downside Deviation
Information Ratio:	Active Return / Tracking Error

Glossary of Key Finance Terms

Term	Definition
Alpha	Excess return above benchmark (risk-adjusted)
Arbitrage	Risk-free profit from price discrepancies
Asset Allocation	Distribution of investments across asset classes
Beta	Sensitivity of asset return to market return
Bid-Ask Spread	Difference between buy and sell price
Book Value	Net asset value per share (Assets - Liabilities) / Shares
Bull Market	Sustained rise in asset prices (> 20% gain from trough)
Bear Market	Sustained fall in asset prices (> 20% decline from peak)
CapEx	Capital Expenditure — investment in long-term assets
CAGR	Compound Annual Growth Rate
Correlation	Statistical relationship between two variables (-1 to +1)
Coupon	Periodic interest payment on a bond
Default	Failure to meet debt obligations
Deleveraging	Reducing debt levels
Derivative	Contract whose value depends on an underlying asset
Discount Rate	Rate used to compute present value of future cash flows
Diversification	Spreading investments to reduce unsystematic risk
Dividend	Distribution of company profits to shareholders
EBIT	Earnings Before Interest and Taxes
EBITDA	Earnings Before Interest, Taxes, Depreciation, Amortization
Enterprise Value	Market Cap + Debt - Cash
EPS	Earnings Per Share
Equity	Ownership stake in a company
Fixed Income	Debt instruments with predetermined cash flows
Free Float	Shares available for public trading (not held by insiders)

Goodwill	Premium paid in an acquisition above book value
Hedge	Position taken to reduce risk in another position
IPO	Initial Public Offering — first public sale of company shares
Leverage	Using borrowed funds to amplify returns (and risk)
Liquidity	Ease of converting an asset to cash without price impact
Margin of Safety	Discount to intrinsic value; Benjamin Graham's principle
Mark-to-Market	Revaluing assets at current market prices
Maturity	Date when bond principal is repaid
NAV	Net Asset Value (total assets - liabilities) / units
Par Value	Face value of a bond or stock
Portfolio	Collection of investments held by an investor
Premium	Price above face/par/intrinsic value
Short Selling	Borrowing and selling an asset, hoping to buy back lower
Spread	Difference in yield between two bonds
Sunk Cost	Past cost that cannot be recovered
Systematic Risk	Market-wide risk (cannot be diversified)
Unsystematic Risk	Company-specific risk (can be diversified)
Volatility	Measure of price fluctuation (standard deviation of returns)
Working Capital	Current Assets - Current Liabilities
Yield	Return on investment (interest or dividend income / price)
YTM	Yield to Maturity — total return if bond held to maturity

This document was compiled as a comprehensive finance reference for educational, analytical, and RAG (Retrieval-Augmented Generation) testing purposes. All formulas, data, and classifications reflect standard financial theory and practice as of 2024. For actual investment decisions, consult a SEBI-registered financial advisor.

